AKSHAT ATUL BHARGAVA

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PROFESSIONAL SUMMARY

NUS Data Science graduate student with 3 years of industry experience building scalable data-driven systems at Mastercard. Skilled in Python, SQL, and machine learning with hands-on experience in large-scale data pipelines (PySpark, 100M+ records), anomaly detection, statistical modelling, and cross-functional analytics. Seeking to apply data science skills in a high-impact internship role.

WORK EXPERIENCE

Software Engineer II | Mastercard, Pune *Mar 2024 – Jul 2025*

- Built Datadog monitoring dashboards and anomaly detection workflows across 5+ microservices, analyzing large-scale transaction telemetry — **reduced incident response time by ~40%.**
- Designed load/spike test experiments under varying traffic distributions; informed **data-driven capacity planning and performance benchmarking.**
- Developed 5+ scalable backend microservices with 100+ automated test cases (near-100% coverage); reduced post-deployment bugs by 20% via structured code reviews across 4+ country teams.

Software Engineer I | Mastercard, Pune *Jul 2022 – Mar 2024*

- Optimized legacy API architecture using profiling and metrics dashboards — **improved response time by 30%, cut technical debt by 50%.**
- Led backend feature development for 4 major releases across full SDLC; automated API & regression testing (Postman) to boost coverage and streamline release cycles.

PROJECTS

Large-Scale Weather Prediction — Apache Spark MLlib | *Python, PySpark, Spark MLlib, GCP Dataproc, Random Forest*
GitHub: akshatbhargava2906/machine-learning-on-weather-data

- Ingested & cleaned **100M+ hourly NOAA weather records** on GCP Dataproc; preprocessing pipeline reduced dataset to ~18M high-quality records for distributed training.
- Trained Random Forest Regressor (Spark MLlib) achieving **RMSE 4.03°C, R^2 = 0.89**; feature importance analysis identified dew point, spatial & temporal features as key predictors.

One-Shot Face Recognition (Siamese Neural Network) | *Python, TensorFlow, Keras, OpenCV*
GitHub: akshatbhargava2906/one_shot_face_recognition

- Built a one-shot face recognition system using a **Siamese Neural Network with shared CNN embeddings** (4 Conv2D blocks -> Dense 4096) and a **custom L1 distance layer** for image pair similarity scoring.
- Collected and curated a labelled dataset using webcam capture (anchor/positive) and the LFW dataset (negatives).

Movie Recommender System – Apache Spark MLlib | *Python, PySpark, GCP Dataproc*
GitHub: akshatbhargava2906/movie-recommender-system

- Built a scalable collaborative filtering system on the **MovieLens dataset (20M ratings)** to generate personalized movie recommendations for **138K users.**
- Implemented ALS matrix factorization with K-Means user/movie clustering; achieved **RMSE: 0.8110 and MAE: 0.6349 on a 30% held-out test set**

EDUCATION

M.Sc. Data Science & Machine Learning *Ongoing – Dec 2026*
National University of Singapore (NUS) | Courses: Scalable Distributed Systems, Big Data Analytics, Machine Learning, Data Management

B.E. Electronics & Telecommunication *CGPA: 9.01 / 10*
Pune Institute of Computer Technology | Courses: Data Structures, Artificial Intelligence, OOP, Business Management

SKILLS

Programming & Querying: Python, SQL, Java
Data Science & ML: Pandas, NumPy, Scikit-learn, PySpark, Feature Engineering, EDA, Anomaly Detection, Model Evaluation Metrics.
Data Engineering: Data Pipeline Design, Spark MLlib, GCP Dataproc, SQL Databases, REST APIs
Visualization & Monitoring: Datadog, Jupyter, Matplotlib, Seaborn, Power BI / Tableau (familiar)
MLOps & Dev Tools: Git, Jenkins, Unit Testing, Postman, Scalable Systems

INTERESTS

Sports: Badminton, Cricket, Table Tennis | **Personal:** Guitar (Trinity College London, Grade 6), Reading, Formula 1